

Does Stratified Sampling Matter? Evidence from Synthetic, MNIST, and CIFAR-10

1. Research Question

Does stratified sampling affect model performance differently depending on task difficulty and data size?

This study compares three datasets spanning a difficulty spectrum: synthetic (easy), MNIST (easy-medium), and CIFAR-10 (hard).

2. Experimental Setup

	Synthetic	MNIST	CIFAR-10
Source	<code>make_classification</code> (20 feat.)	digit 1 vs 7 (784 pixels)	airplane vs cat (3072 pixels)
Difficulty	Easy	Easy-Medium	Hard
Imbalance	9:1	9:1 (subsampled)	9:1 (subsampled)
Data sizes	50, 500, 5K, 50K	100, 500, 5K, 15,170	100, 500, 5K, 12,000
Models	DT, KNN, LR, SVM, MLP	Same	Same
Repetitions	30 seeds, <code>model_seed=0</code>	Same	Same

All models use flattened features + StandardScaler. MLP: 3-layer, 50 epochs, fixed model initialization seed.

3. Results

3.1 Paired t-test Summary (All significant results, $p < 0.05$)

Dataset	n	Model	t-stat	p-value	Winner
Synthetic	50	KNN	-2.363	0.0250	Random
Synthetic	50	SVM	-3.032	0.0051	Random
Synthetic	5,000	LR	-2.211	0.0351	Random

Dataset	n	Model	t-stat	p-value	Winner
MNIST	100	LR	+2.291	0.0294	Stratified
MNIST	500	KNN	-2.567	0.0157	Random
CIFAR-10	100	LR	+2.130	0.0418	Stratified
CIFAR-10	500	SVM	-2.321	0.0275	Random
CIFAR-10	12,000	SVM	+2.484	0.0190	Stratified

8 significant out of 60 tests (13%). Direction is mixed: 5 favor Random, 3 favor Stratified.

3.2 Significant Results by Dataset

Dataset	Total sig.	Stratified better	Random better
Synthetic	3	0	3
MNIST	2	1	1
CIFAR-10	3	2	1

Pattern: As task difficulty increases, Stratified becomes more beneficial.

3.3 Diff Mean Direction at Largest Data Size

Model	Synthetic (n=50K)	MNIST (n=15K)	CIFAR-10 (n=12K)
DT	+0.0010	+0.0003	+0.0013
KNN	+0.0006	+0.0002	+0.0023
LR	+0.0003	+0.0003	+0.0026
SVM	+0.0007	+0.0002	+0.0039
MLP	+0.0008	-0.0000	+0.0031

At the largest sample size, all three datasets show **positive** diff means (Stratified slightly better). CIFAR-10 shows the largest magnitudes — on hard tasks, the benefit of Stratified persists even at large n.

3.4 Ceiling Effect Comparison

Average F1 across both strategies at n=500:

Model	Synthetic	MNIST	CIFAR-10
MLP	0.858	0.978	0.718

Model	Synthetic	MNIST	CIFAR-10
LR	0.770	0.980	0.672
SVM	0.798	0.912	0.500
DT	0.718	0.942	0.646
KNN	0.659	0.949	0.557

- **Synthetic**: MLP leads, ceiling effect present
- **MNIST**: LR and MLP both at 0.98 — extreme ceiling, barely any room for split strategy to matter
- **CIFAR-10**: All models well below 0.8 — **no ceiling effect**, plenty of room for variation

Which model has smallest diff_std at n=500?

Dataset	Smallest diff_std	Model
Synthetic	0.0773	MLP
MNIST	0.0275	MLP
CIFAR-10	0.0724	SVM

On CIFAR-10 (no ceiling), MLP is NOT the most stable — SVM is. This confirms ceiling effect drove the synthetic/MNIST results.

3.5 Independent F1 Std at Small n (n=100)

Ratio = Stratified std / Random std. Values < 1.0 mean Stratified reduces variance.

Model	MNIST ratio	CIFAR-10 ratio
DT	0.65	0.72
KNN	0.72	0.80
LR	0.41	1.08
SVM	0.50	0.00 (degenerate)
MLP	0.53	0.81

At small n on real data, Stratified consistently reduces F1 variance — ratios are mostly well below 1.0. This pattern is much clearer than on synthetic data.

4. Findings

Finding 1: Task difficulty determines whether Stratified matters

Difficulty	Dataset	Stratified effect
Easy	Synthetic	No benefit; Random sometimes better
Easy-Medium	MNIST	Small benefit at n=100 (LR significant); otherwise negligible
Hard	CIFAR-10	Consistent positive trend; SVM significantly benefits at n=12K

Why? On hard tasks, class boundaries overlap more. A skewed test set (from random split) can disproportionately misrepresent the minority class boundary region, leading to unreliable evaluation.

Finding 2: Stratified reliably reduces variance on small real data

At n=100 on both MNIST and CIFAR-10, most models show Strat/Rand std ratio < 1.0 (0.41 to 0.81). This was NOT visible on synthetic data. **Stratified's primary value on real data is stabilization, not performance improvement.**

Finding 3: Ceiling effect explains MLP's apparent stability

- MNIST (easy): MLP F1 = 0.978 at n=500. Diff_std = 0.028. Smallest. But there's almost no room to vary.
- CIFAR-10 (hard): MLP F1 = 0.718 at n=500. Diff_std = 0.090. **4th out of 5** — not the most stable.
- The "MLP is less sensitive" narrative is an artifact of high baseline performance, not an inherent property of neural networks.

Finding 4: SVM is the most split-sensitive model — and the most affected by task difficulty

- SVM produced the most significant t-test results across all datasets (3 out of 8)
- On easy data: Stratified hurts SVM (forces degenerate minority splits at small n)
- On hard data: Stratified helps SVM significantly (p=.019 at CIFAR-10 n=12K)
- SVM's kernel-based decision boundary is especially sensitive to training class distribution

Finding 5: Data size remains the dominant factor

All three datasets show the same pattern:

From → To	Diff_std reduction
Synthetic n=500 → 50K	~25x
MNIST n=500 → 15K	~18x
CIFAR-10 n=500 → 12K	~8x

The reduction is smaller on CIFAR-10 — harder tasks retain more sensitivity to split strategy even at larger n.

5. Practical Recommendations

Scenario	Recommendation	Evidence
Easy classification	Stratified optional	Synthetic + MNIST: no significant benefit at $n \geq 500$
Hard classification	Use Stratified	CIFAR-10: consistent positive effect, significant for SVM at $n = 12K$
Small data ($n < 200$) on real tasks	Use Stratified	MNIST + CIFAR-10: reliably reduces F1 variance (ratio 0.4-0.8)
Large data ($n > 10K$)	Stratified has slight benefit on hard tasks, no downside	All datasets
SVM specifically	Always use Stratified on non-trivial tasks	SVM most sensitive across all experiments

6. Limitations

1. Only two real datasets tested. More domains needed (tabular medical data, NLP, etc.)
2. MLP with flattened pixels is not representative of modern DL (CNN/Transformer)
3. Only binary classification with 9:1 imbalance
4. No cross-validation or mini-batch-level stratification tested
5. Single data generation seed per size (synthetic) / single subsample (real)

7. Conclusion

Stratified sampling matters — but how much depends on task difficulty.

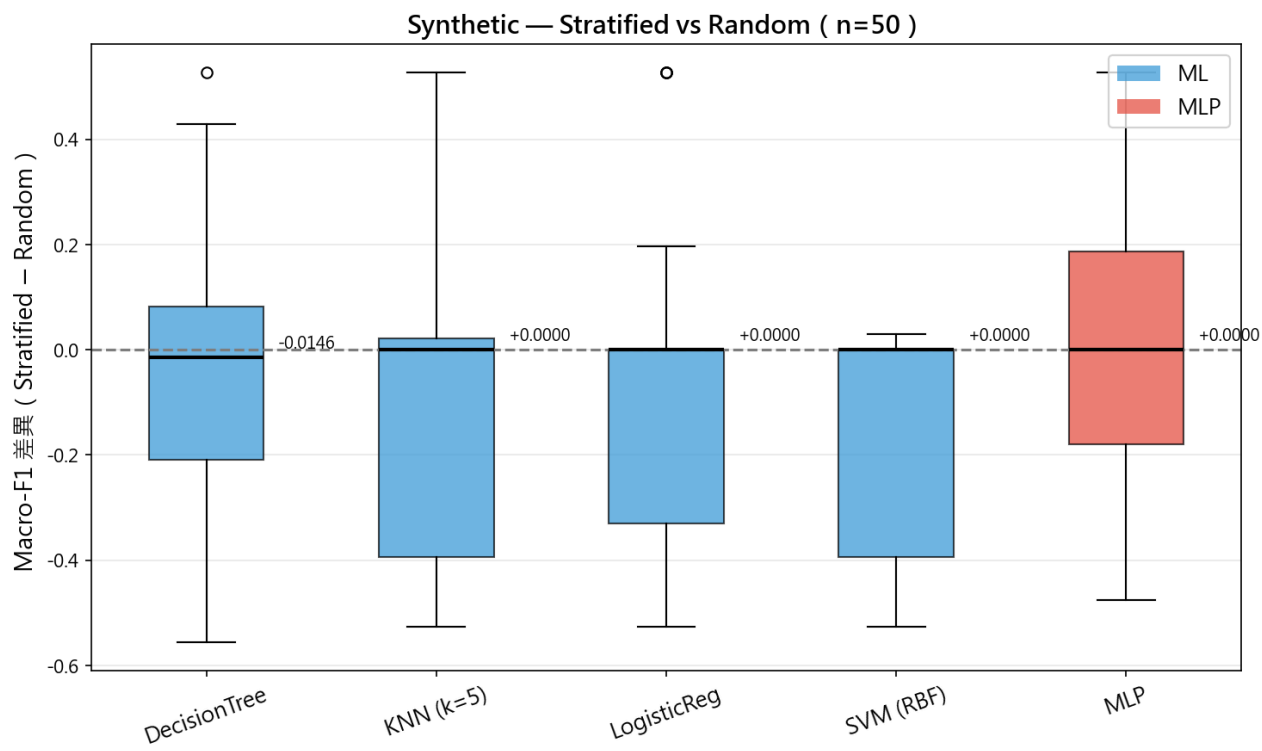
On easy tasks (synthetic, MNIST digit 1 vs 7), stratified and random splits produce nearly identical results. On hard tasks (CIFAR-10 airplane vs cat), stratified sampling provides a small but consistent benefit, particularly for SVM and at both small and large sample sizes.

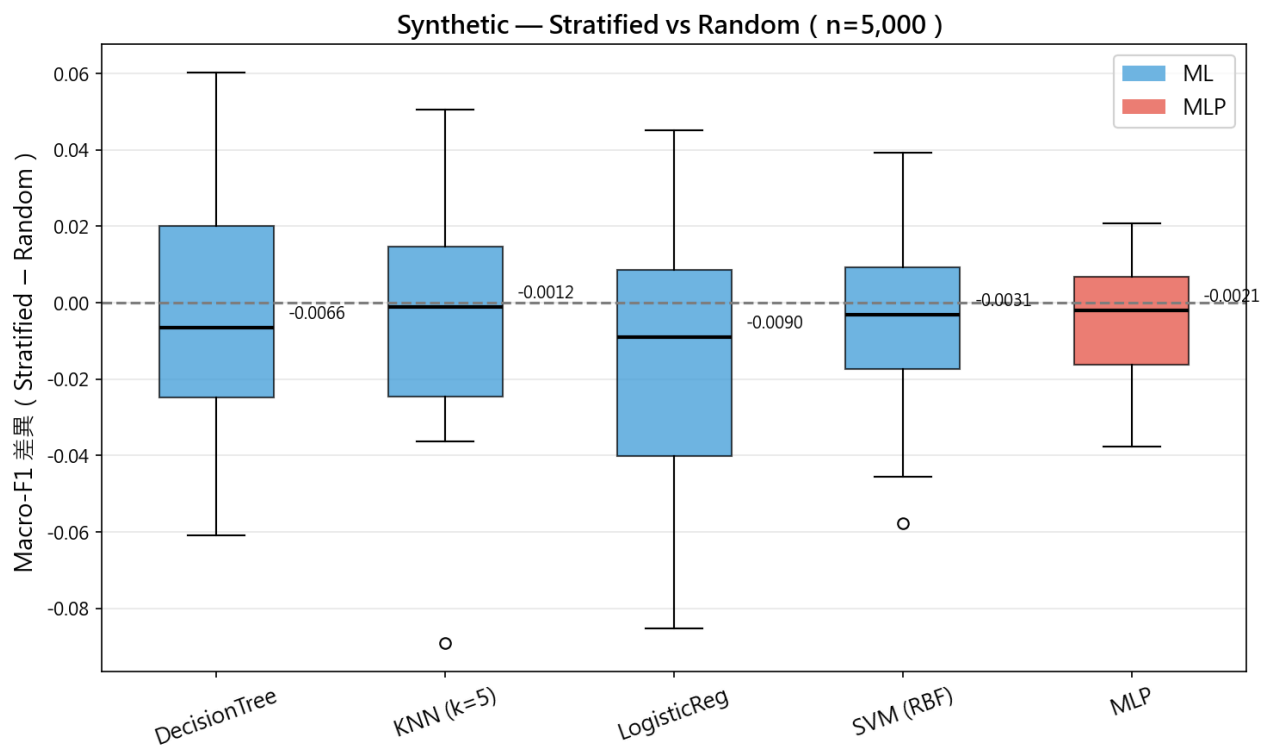
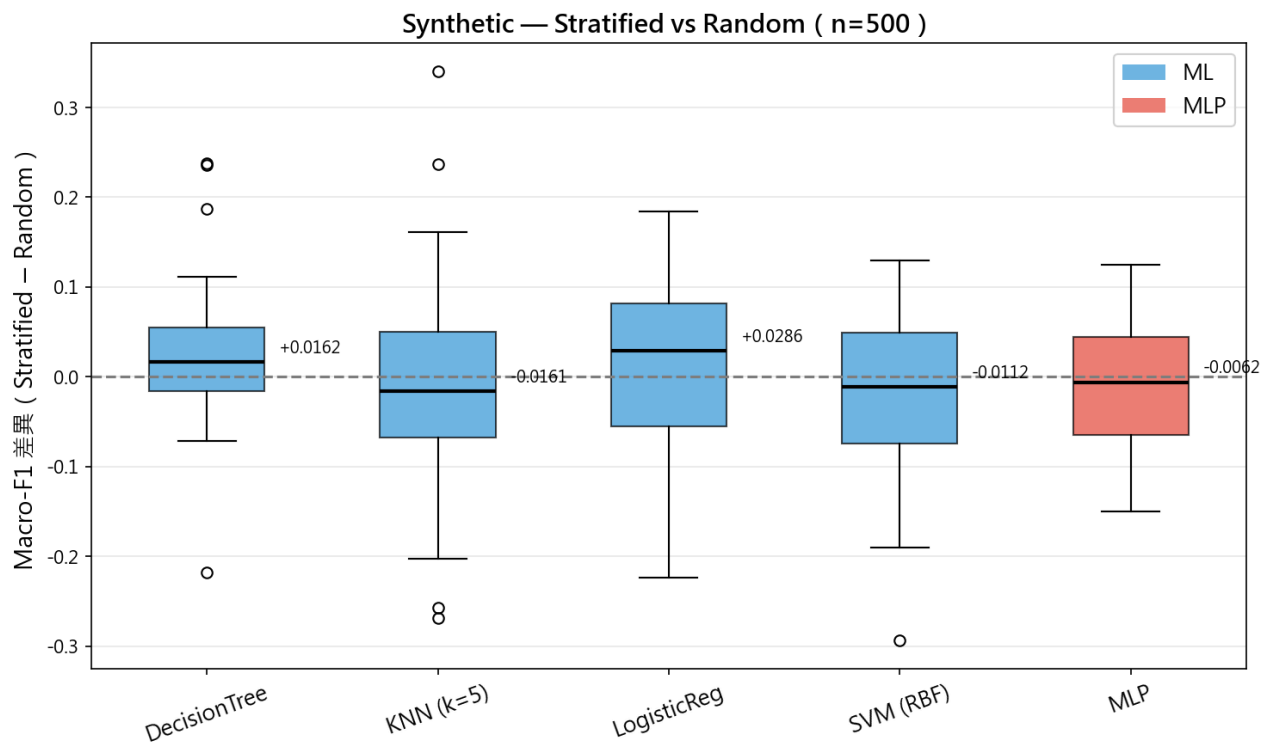
The common claim that "MLP/DL is inherently less sensitive to split strategy" is not supported. MLP's apparent stability is explained by ceiling effect: when a model already achieves high F1, there is less room for the split strategy to make a difference. On hard tasks where no model dominates, MLP shows comparable or greater sensitivity than SVM.

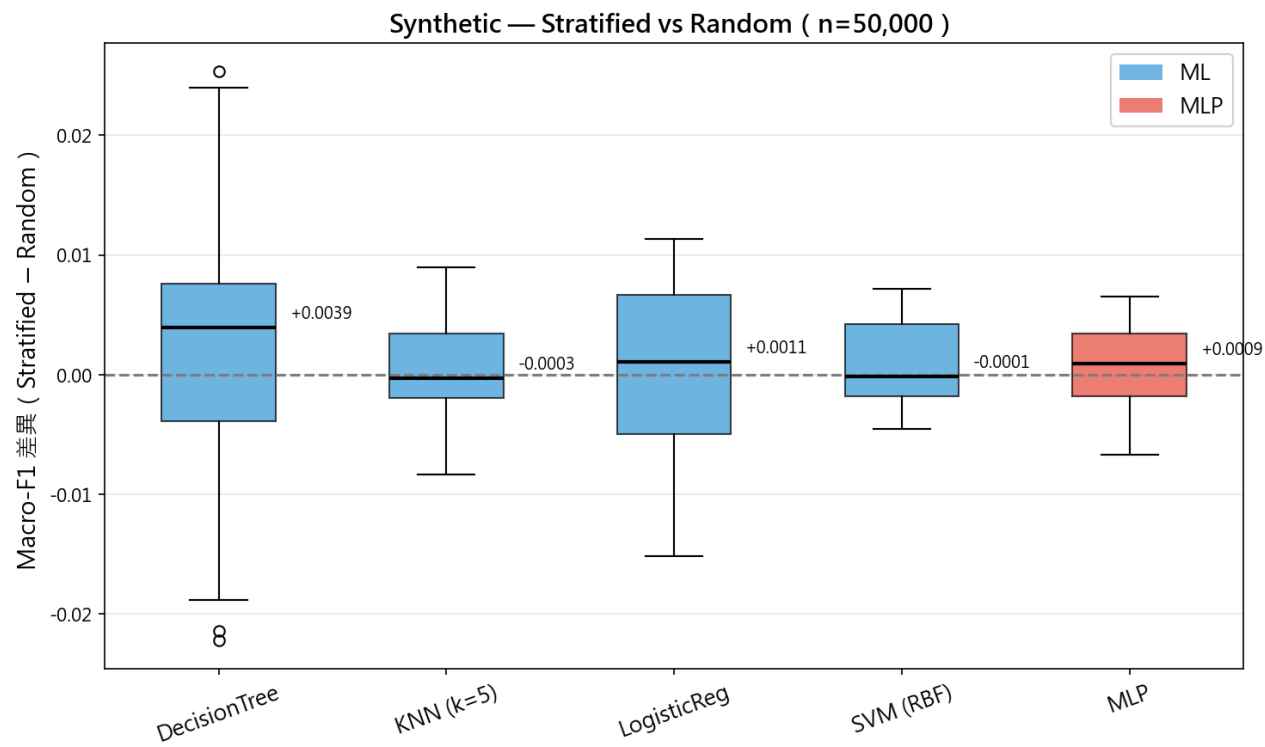
The strongest practical recommendation from this study: always use stratified sampling when the classification task is non-trivial. The cost is negligible, and the benefit — particularly variance reduction on small datasets — is real and measurable.

Appendix: Figures

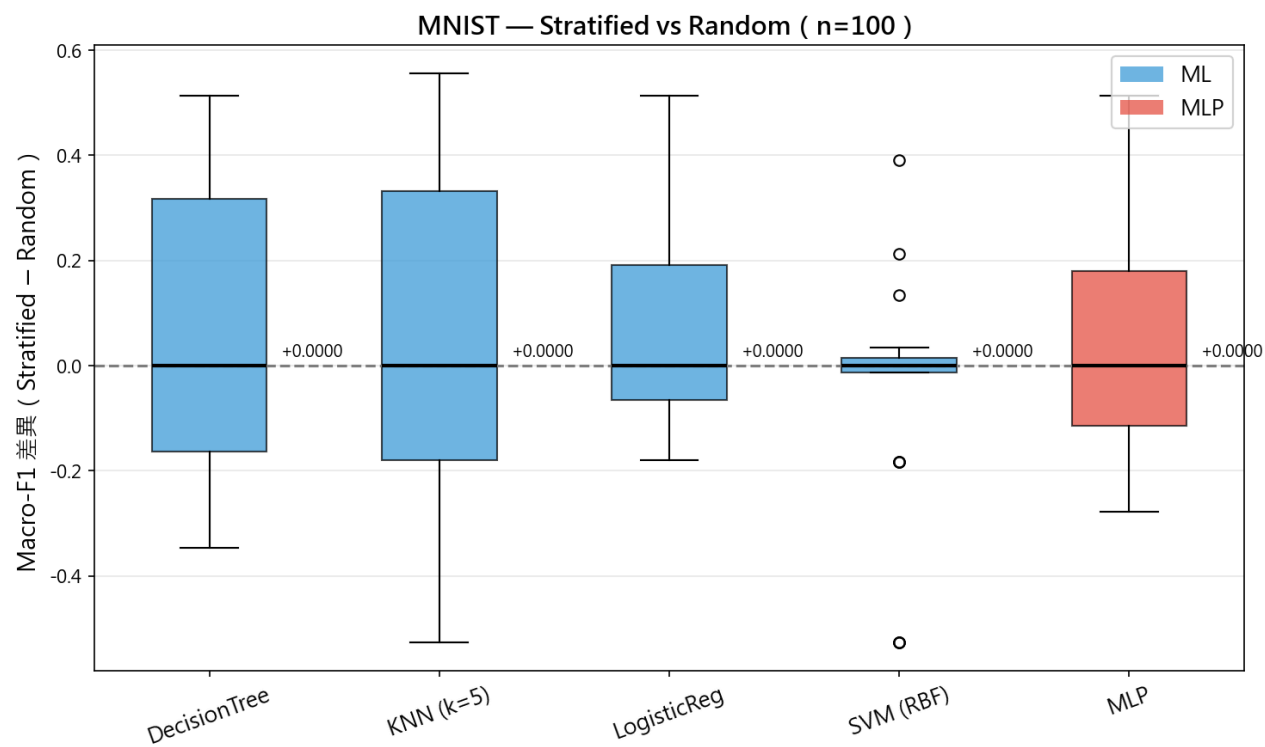
Synthetic

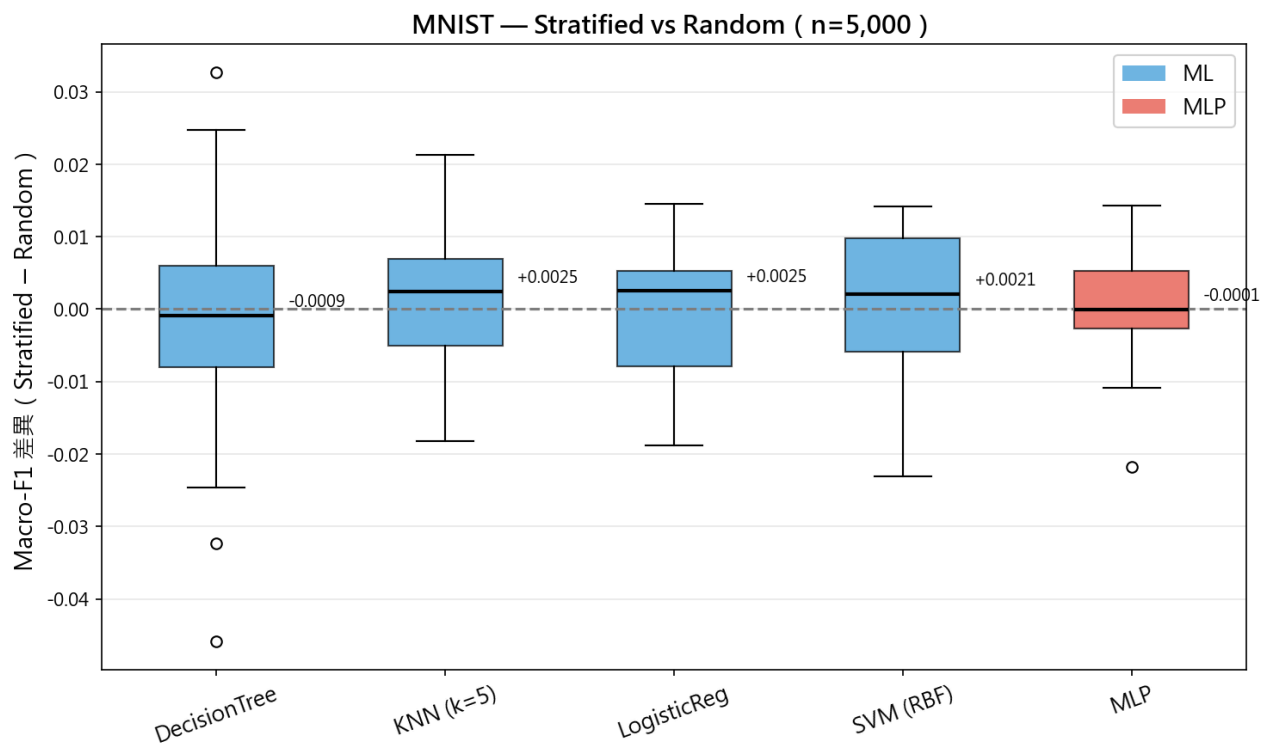
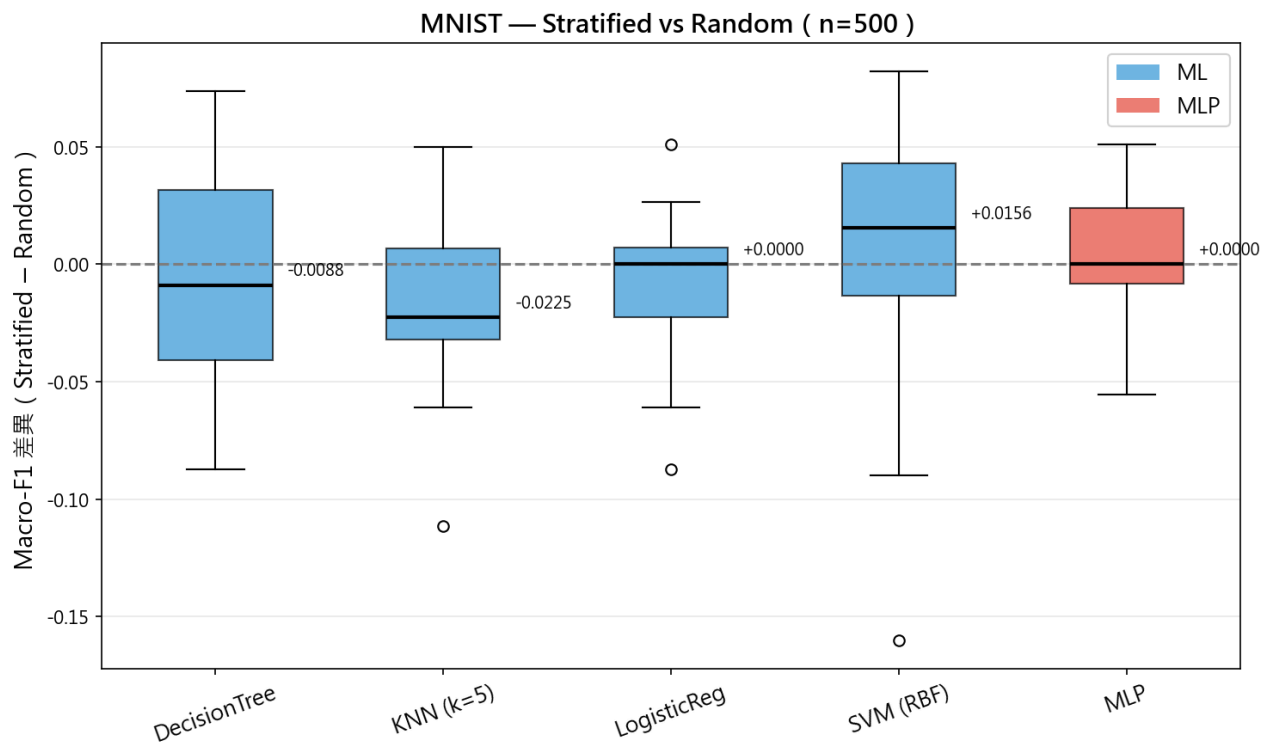


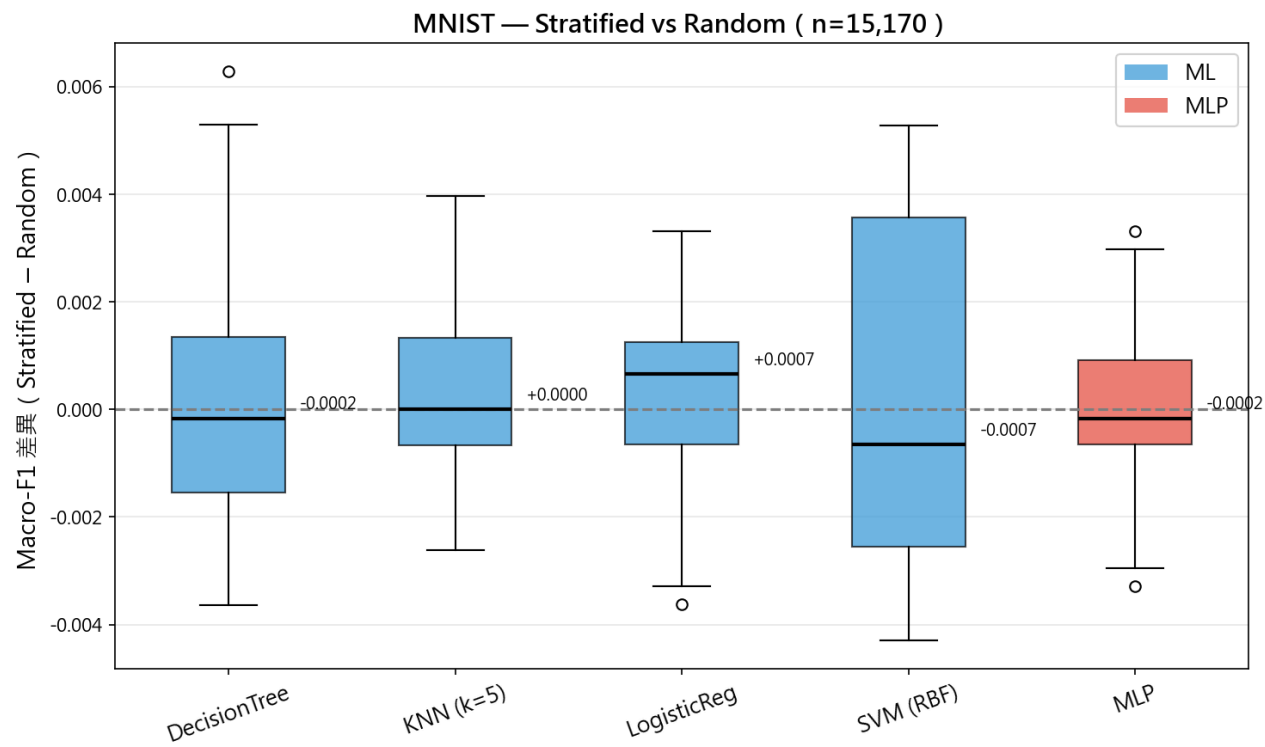




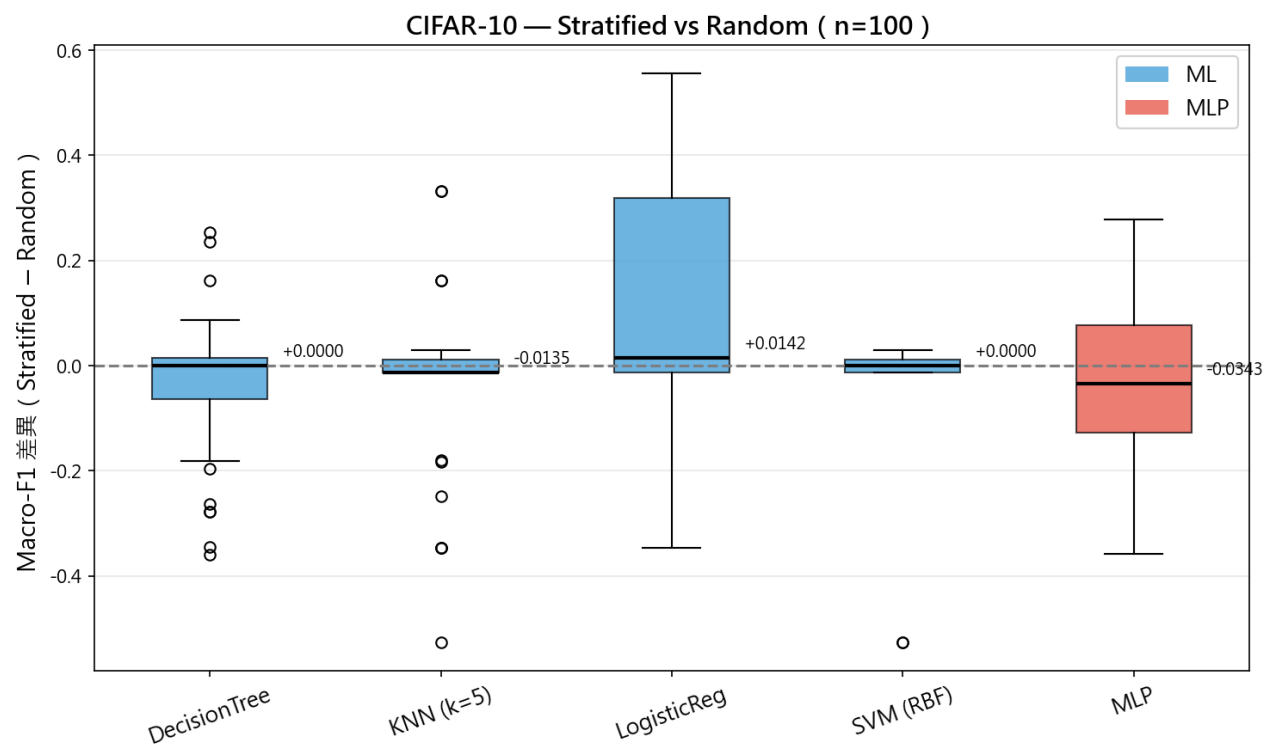
MNIST

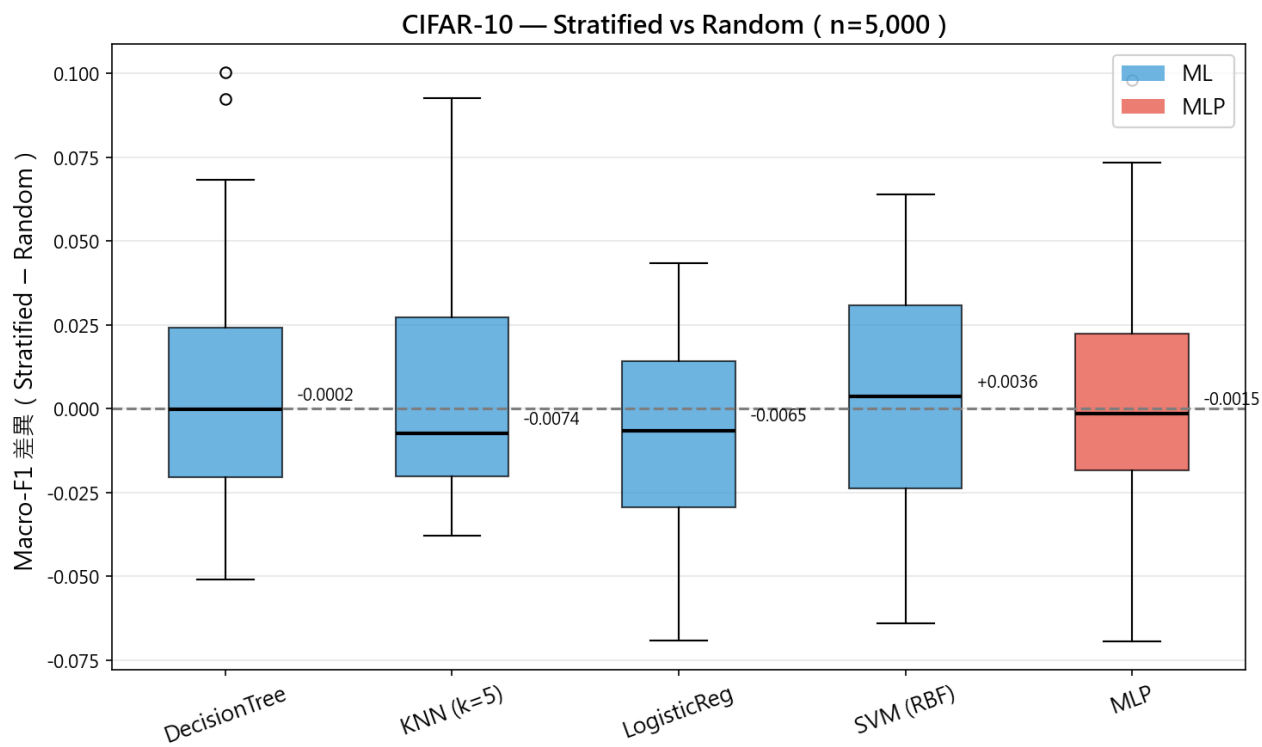
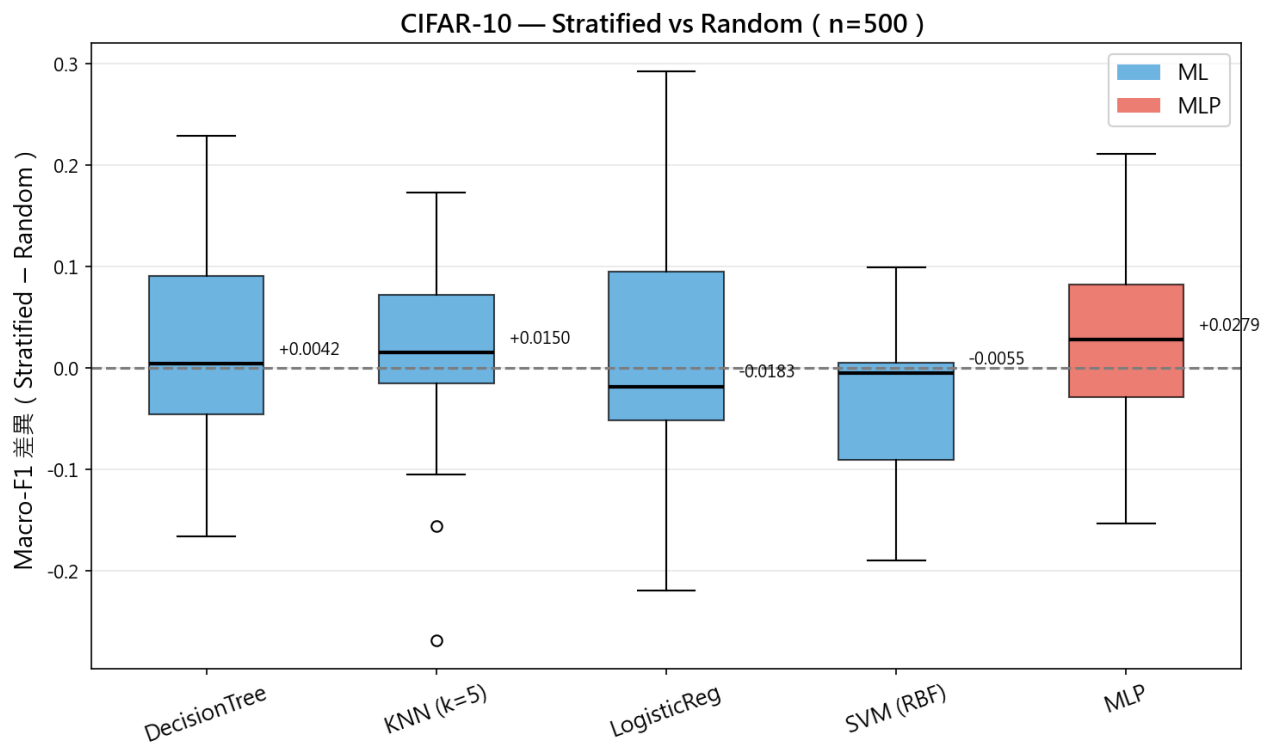


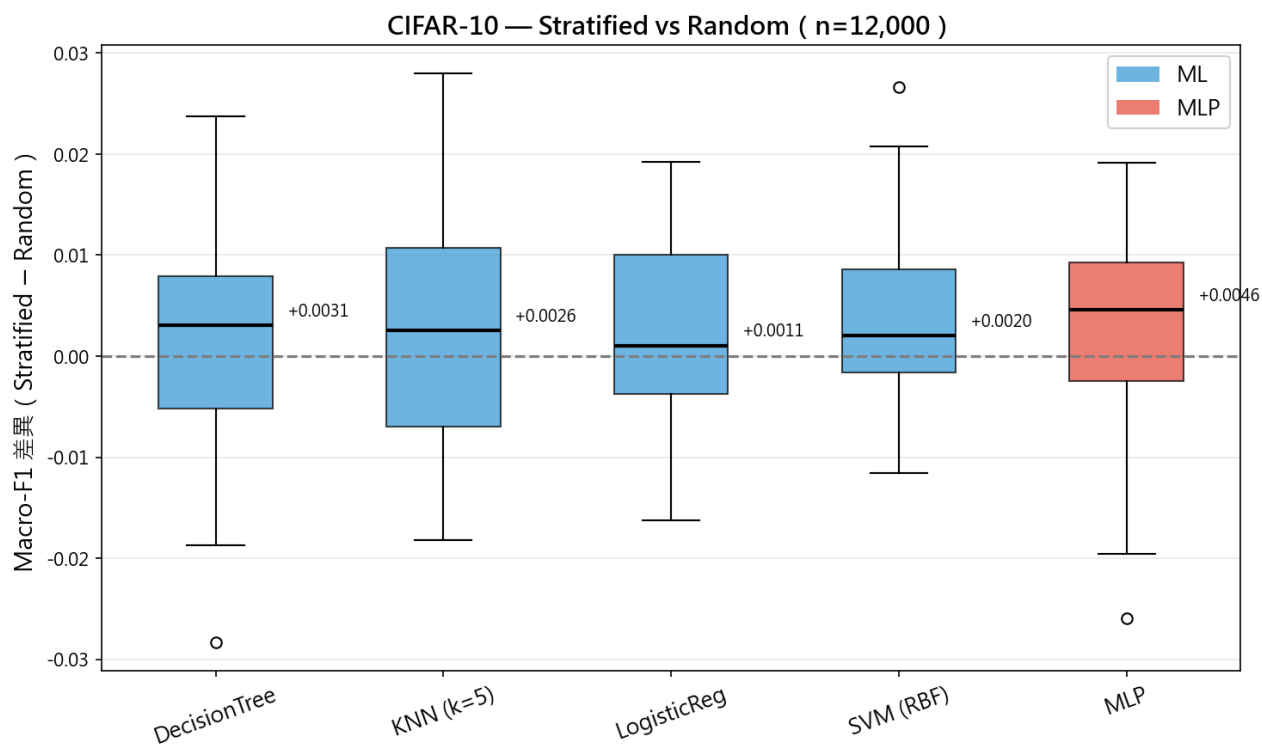




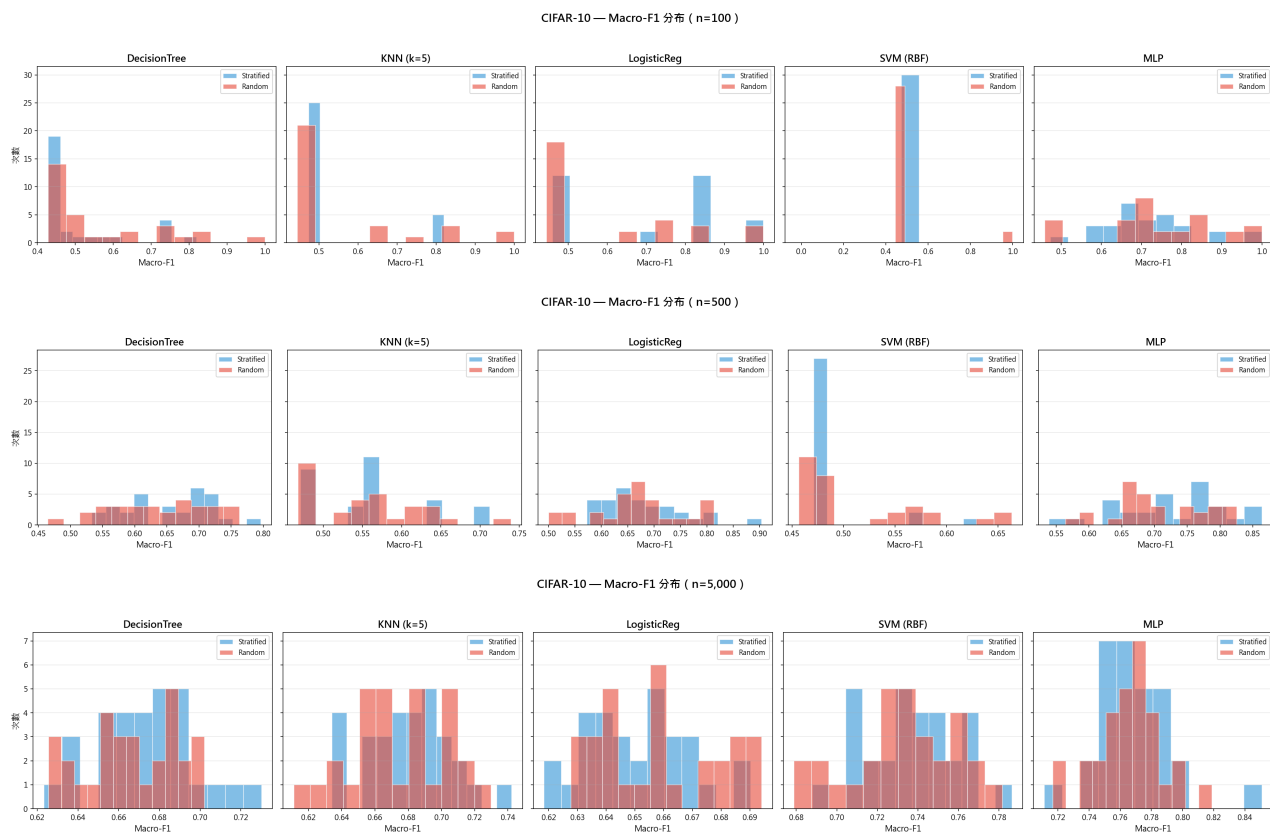
CIFAR-10

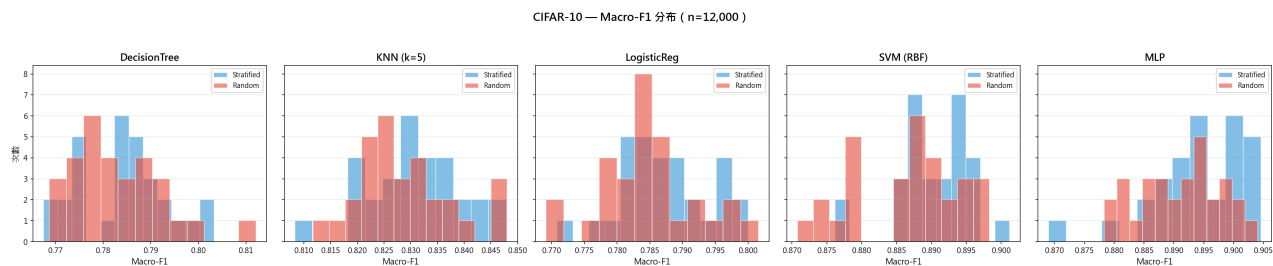






F1 Distributions (CIFAR-10, hardest task)





Test Set Minority Ratio (CIFAR-10)

